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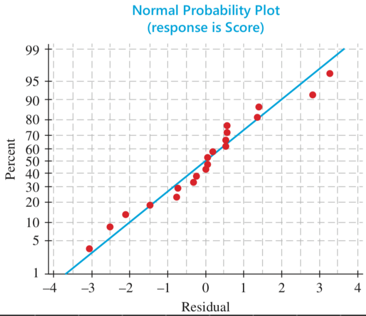
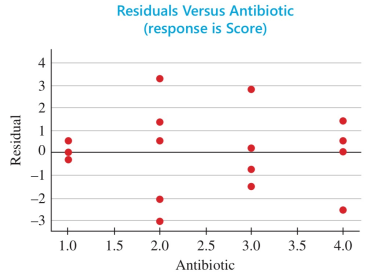
10. Good Tasting Medicine DATA SET DS1528 A voluntary sample of healthy children was used in a study to assess their reactions to the taste of four antibiotics.[7] The children's response was measured on a visual scale using faces, from sad (low score) to happy (high score). The minimum score was 0 and the maximum was 10. For the accompanying data (based on the results of the study), each of five children was asked to taste each of four antibiotics and rate them using the visual (faces) scale from 0 to 10.

	Antibiotic			
Child	1	2	3	4
1	4.8	2.2	6.8	6.2
2	8.1	9.2	6.6	9.6
3	5.0	2.6	3.6	6.5
4	7.9	9.4	5.3	8.5
5	3.9	7.4	2.1	2.0

a. What design is used in collecting these data?

b. The normal probability plot of the residuals as well as a plot of residuals versus antibiotics are shown below. Do the usual analysis of variance assumptions appear to be satisfied?





c. Use the appropriate nonparametric test to test for differences in the distributions of responses to the tastes of the four antibiotics.

a) Randomized Complete Block Design
- Treatments : Antibiotics
- Block : Child

b) The usual ANOVA assumptions do not appear to be fully satisfied especially the equal variance assumption. Therefore, a nonparametric test is appropriate.

c)

	Antibiotic			
Child	1	2	3	4
1	4.8 2	2.2 1	6.8 4	6.2 3
2	8.1 2	9.2 3	6.6 1	9.6 4
3	5.0 3	2.6 1	3.6 2	6.5 4
4	7.9 2	9.4 4	5.3 1	8.5 3
5	3.9 3	7.4 4	2.1 2	2.0 1
Rank Sum	12	13	10	15

H_0: The distributions of responses are the same for the antibiotics.

H_a: At least one antibiotic has different distributions of responses

$n=5$ $k=4$, Rank sums : 12, 13, 10, 15

$$Q = \frac{12}{nk(k+1)} \sum R_j^2 - 3n(k+1)$$

$$= \frac{12}{5(4)(5)} (12^2 + 13^2 + 10^2 + 15^2) - 3(5)(5)$$

$= 1.56$

$df = k-1$
$= 3$

$\chi^2_{3, 0.05} = 7.8147$

$\because Q = 1.56 < \chi^2_{3, 0.05} = 7.8147$

$\therefore$ Non-reject H_0

11. Yield of Wheat Exercise 9 (Chapter 11 Review) presented an analysis of variance of the yields of five different varieties of wheat, observed on one plot each at each of six different locations. The data from this randomized block design are listed here:

DATA SET DS1529

	Location						
Variety	**1**	**2**	**3**	**4**	**5**	**6**	Rank Sum
A	35.3 4	31.0 2	32.7 3	36.8 3	37.2 3	33.1 3	18
B	30.7 1	32.2 3	31.4 2	31.7 2	35.0 2	32.7 2	12
C	38.2 5	33.4 4	33.6 4	37.1 4	37.3 4	38.2 5	26
D	34.9 3	36.1 5	35.2 5	38.3 5	40.2 5	36.0 4	27
E	32.4 2	28.9 1	29.2 1	30.7 1	33.9 1	32.1 1	7

a. Use the Friedman F_r-test to determine whether the data provide sufficient evidence to indicate a difference in the yields for the five different varieties of wheat. Test using $\alpha = .05$.

b. When the analysis of variance was performed in Chapter 11, the ANOVA F-test produced the test statistic $F = 18.61$ with p-value $= .000$. How do these results compare with results of the Friedman F_r-test in part a? Explain.

a) H_0 : The distributions of yields are the same for the five varieties

H_a : At least one variety has a different yield distribution

$b = 6$, $k = 5$

$$Q = \frac{12}{bk(k+1)} \sum R_j^2 - 3b(k+1)$$

$$= \frac{12}{6(5)(6)} (18^2 + 12^2 + 26^2 + 27^2 + 7^2) - 3(6)(6)$$

$= 20.133$

$k - 1 = 4$

$\chi^2_{4,0.05} = 9.4877$

$\because Q = 20.133 > \chi^2_{4,0.05} = 9.4877$

$\therefore$ Reject H_0

b) $F = 18.61$, p-value $= 0.00$

$\therefore$ Both test lead to the same conclusion : the five wheat varieties do not have the same yield distribution

DATA SET

DS1530

11. Rating Political Candidates A political scientist is studying the relationship between the voter image of a conservative political candidate and the distance (in kilometers) between the residences of the voter and the candidate. Each of 12 voters rated the candidate on a scale of 1–20.

Voter	Rating		Distance		d	d^2
1	12	7.5	75	3	45	20.25
2	7	4	165	7	-3	9
3	5	3	300	12	-9	81
4	19	12	15	1	11	121
5	17	10	180	8	2	4
6	12	7.5	240	11	-3.5	12.25
7	9	6	120	4	2	4
8	18	11	60	2	9	81
9	3	1	230	10	-9	81
10	8	5	200	9	-4	16
11	15	9	130	5.5	3.5	12.25
12	4	2	130	5.5	-3.5	12.25

a. Calculate Spearman's rank correlation coefficient r_s.

b. Do these data provide sufficient evidence to indicate a negative rank correlation between rating and distance?

a) $H_0: \rho_s = 0$

$H_a: \rho_s < 0$

$n = 12$

$$r_s = 1 - \frac{6\sum d^2}{n(n^2-1)}$$

$$= 1 - \frac{6(454)}{12(144-1)}$$

$$= -0.5874$$

b) $n = 12$, $\alpha = 0.05$

critical value $= 0.497$

$\because r_s = -0.5874 < -0.497$

$\therefore$ Reject H_0

DATA SET

DS1531

12. Competitive Running Is the number of years of competitive running experience related to a runner's distance running performance? The data on nine runners, obtained from a study by Scott Powers and colleagues, are shown in the table:[8]

Runner	Years of Competitive Running	R_x	10-Kilometer Finish Time (minutes)	R_y	$R_x - R_y$	d^2
1	9	7	33.15	1	6	36
2	13	9	33.33	2	7	49
3	5	3.5	33.50	3	0.5	0.25
4	7	6	33.55	4	2	4
5	12	8	33.73	5	3	9
6	6	5	33.86	6	-1	1
7	4	2	33.90	7	-5	25
8	5	3.5	34.15	8	-4.5	20.25
9	3	1	34.90	9	-8	64
						208.5

a. Calculate the rank correlation coefficient between years of competitive running and a runner's finish time in the 10-kilometer race.

b. Do the data provide evidence to indicate a significant rank correlation between the two variables? Test using $\alpha = .05$.

a) $n=9$

$$r_s = 1 - \frac{6\sum d^2}{n(n^2-1)}$$

$$= 1 - \frac{6(208.5)}{9(81-1)}$$

$$= -0.738$$

b) $H_0: \rho_s = 0$

$H_a: \rho_s \neq 0$

$n=9$, $\alpha = 0.05$

$\therefore$ Critical value $= \pm 0.683$

$\because |r_s| = 0.738 > 0.683$

$\therefore$ Reject H_0